ENEL 573 – Computer Networks

Chapter 2: Network Analysis and Queueing Theory

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Two approaches to network design

1- "Build first, worry later" approach

- More ad hoc, less systematic

2- "Analyze first build later" is used extensively for telephone networks, computer networks

- More systematic, optimal, etc.

- A model is a mathematical abstraction: keep only the details that are relevant
- Mathematical modeling is one step that can provide useful approximations

- Mathematical model can be used for:

  1- Evaluate the system performance
    - Average waiting time
    - Throughput
    - Loss probability due to buffer overflow

  2- Improve the system performance

    - Determine the service rate with tolerable waiting time (upgrade the system with more capacity incurs investment cost. But long waiting time would be annoying to users)

    - Provide guaranteed packet loss probability with large enough buffer (How large is enough?)

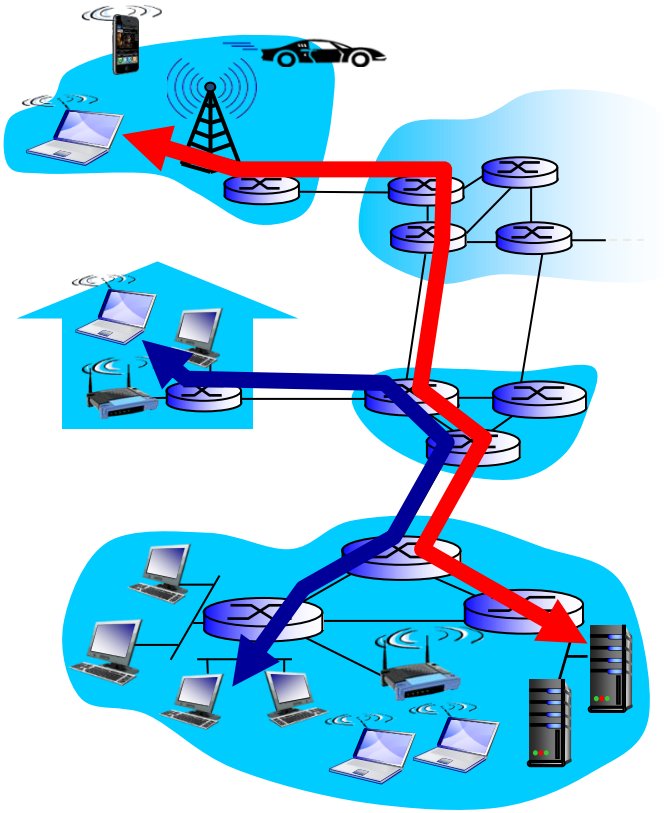
Roadmap

- Randomness in communication networks
- A review of probability
- Queuing theory

Randomness at the source (1/2)

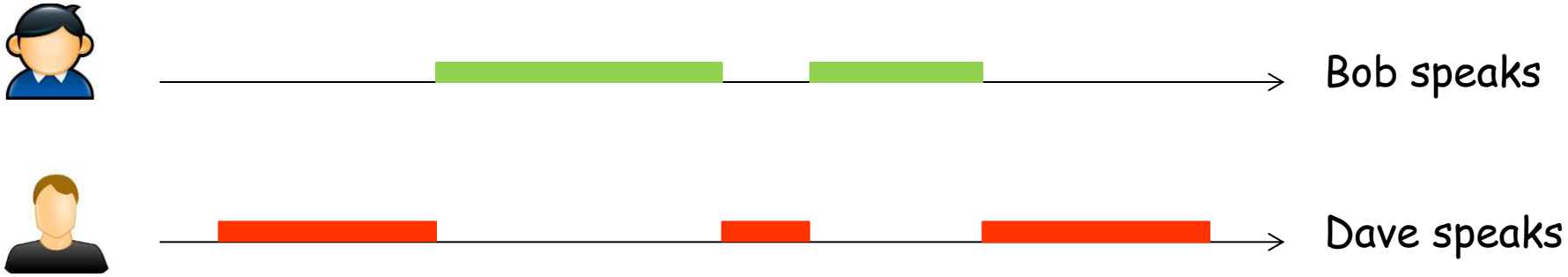
- Communication is full of randomness during the data generation at the source and data transmission at the network

- Random data generation at the source
  - data: users randomly subscribe to download at any time (e.g., web browsing)
  - voice: dynamic changing of data flow based on the interactivity during the conversation
  - video: variable video rate

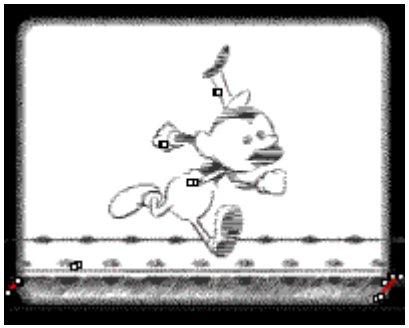


Randomness at the source (2/2)

- Voice communication: interactive communication



- Video: variable length of video frames

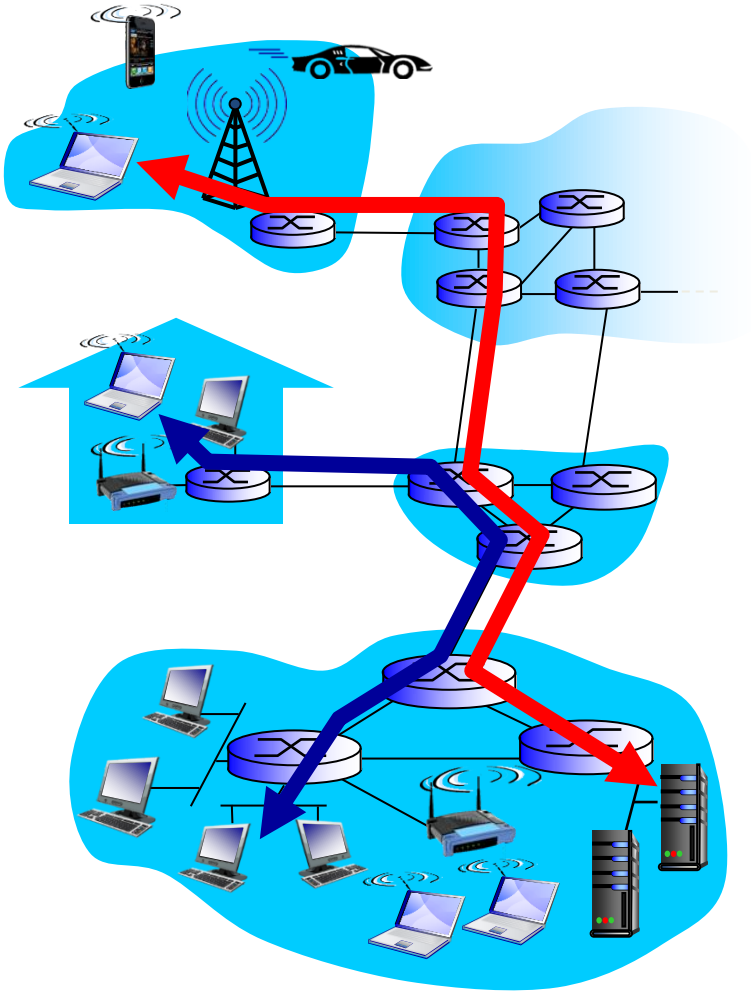




  - Video = moving picture (30 frames/s)
  - picture (frame) size depends on motion in pictures

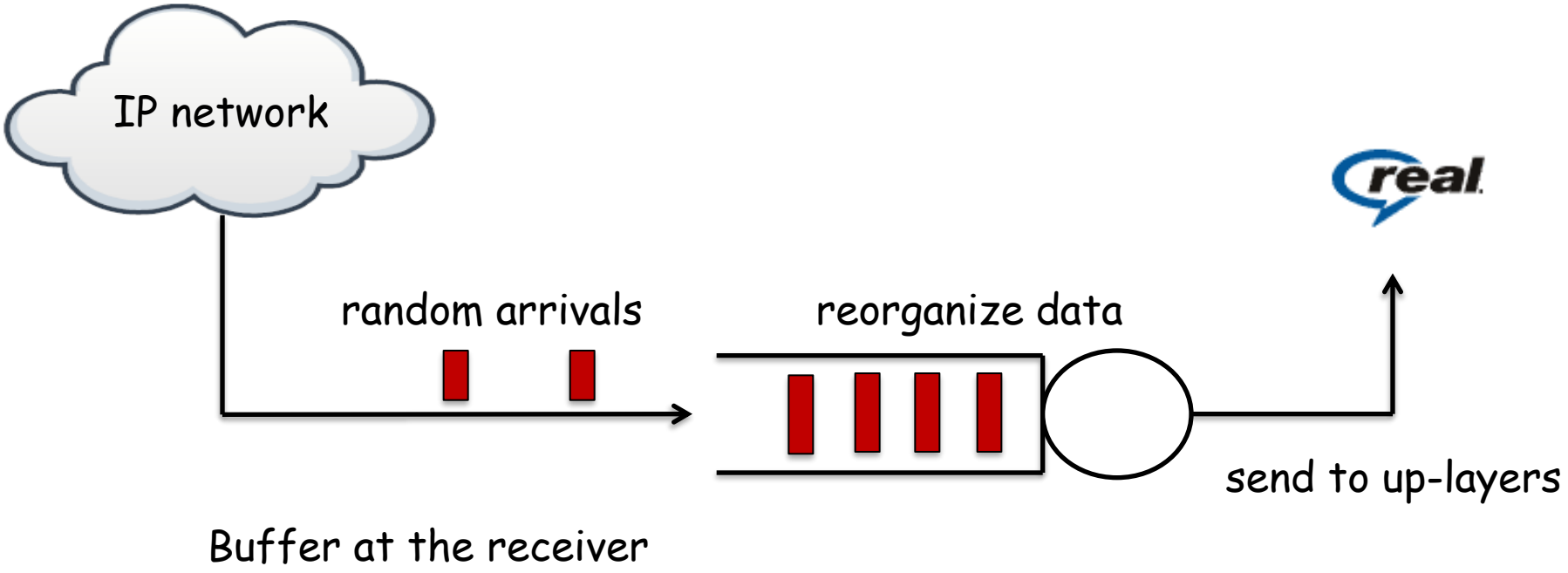
Randomness at the network

- Data transmission encounters random delays and data rate
  - randomly changing physical channels (noise of wireless communication)
  - congestion in the backbone network (statistical multiplexing of data packets)
  - contention of transmission among users in access network



Buffers

- Buffers are used to absorb the randomness of network
  - Sender, routers, receivers all have buffers



Roadmap

- Randomness in communication networks
- A review of probability
- Queuing theory

A review of probability

- Definitions
- Random Variables
- Distribution of RVs
- Statistical Characterizations
- Useful Distributions

Definitions

- Experiment: a physical experiment with an observable outcome
- Random Experiment: an experiment with a non-deterministic outcome (e.g., packet transmission with random lost)
- Sample Space (S): the set of all possible well-defined outcomes of an experiment
- Event (E): a subset of the sample space

Example

Tossing a die. The sample space is $S = \{1, 2, 3, 4, 5, 6\}$. $E = \{2, 4, 6\}$ is an event, which can be described in words as "the number is even".

- Probability of an event E can be defined in terms of its relative frequency as:

$$P(E) = \lim_{n \to \infty} \frac{n(E)}{n}$$

Conditional Probability

- The conditional probability of an event E assuming that an event F occurs, denoted by P (E|F), defined as:

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

- Probability of joint events is:

$$P(A \cap B) = P(A|B)\,P(B)$$

- If A and B are independent

$$P(A \cap B) = P(A)\,P(B)$$

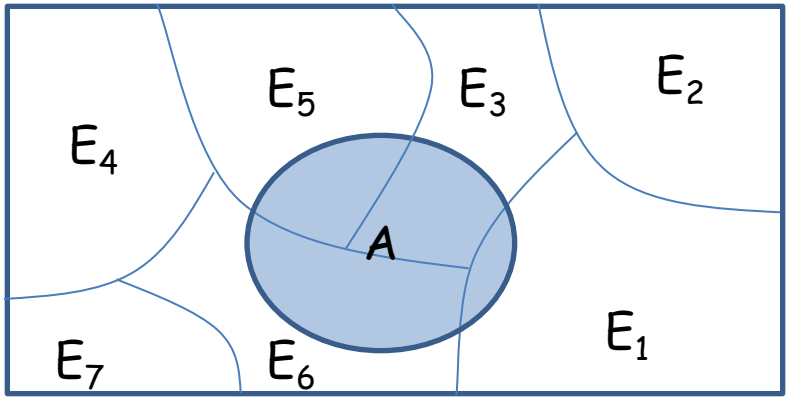
Example

$A = \{\text{Alice goes to bar today}\}, B = \{\text{Bob goes to bar today}\}$. What is $P(A \cap B)$, the probability that both Alice and Bob will go to bar today? Suppose that: 1. $P(B) = 0.5$; and 2. if Bob goes, Alice will go for sure.

Total Probability Theorem

If $\{E_1, E_2, ..., E_n\}$ is a partition of S, that is (mutually disjoint)

$$\bigcup_{i=1}^{n} E_n = S, \text{ and } E_i \cap E_j = \Phi \text{ for } 1 \leq i, j \leq n,$$



A is an arbitrary event in S, the total probability theorem is given as:

$$P(A) = P(A|E_1)\,P(E_1) + P(A|E_2)\,P(E_2) + \cdots + P(A|E_n)\,P(E_n)$$

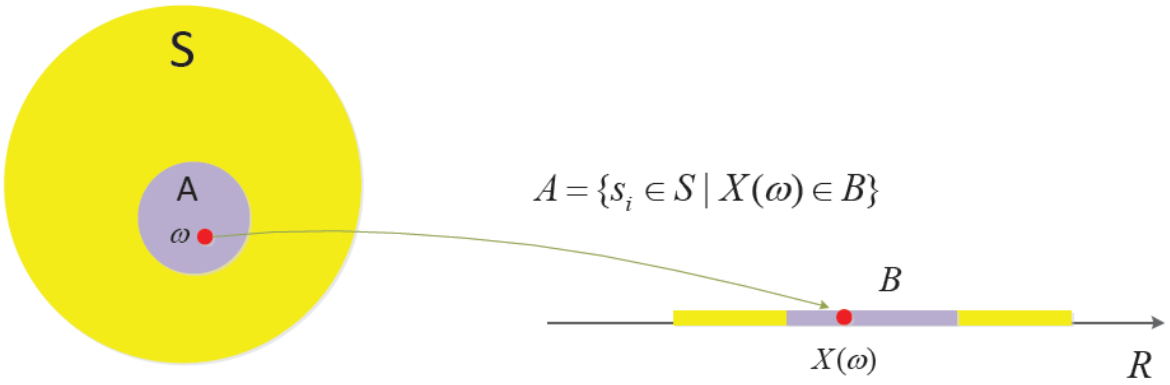
Proof:

$$\begin{aligned} A &= A \cap S &&= A \cap (E_1 \cup E_2 \cup \cdots \cup E_n) \\ &&&= (A \cap E_1) \cup (A \cap E_2) \cup \cdots \cup (A \cap E_n)\,. \end{aligned}$$

$$P(A) = \sum_{i=1}^{n} P(A \cap E_i) = \sum_{i=1}^{n} P(A|E_i)\,P(E_i)$$

Random Variables

- A random variable (r.v.) X(ω) is a function that maps an arbitrary outcome ω of a random event into a real value



Example

- $X = 1$, if a user subscribe to watch "Transformer I" from the Youtube server in the next time slot; otherwise, $X = 0$
- Bandwidth consumed by a user; sojourn time of user connections; contents that users subscribed to download, ...

Distribution of Random Variables

- Cumulative Density Function (CDF)

$$F(x) = P(X \leq x)$$

specifying, for all real $x \in (-\infty, \infty)$, the probability that the r.v (random variable) X is less than or equal to x.

- Probability Density Function (pdf) of a continuous r.v.

$$f(x) = \frac{dF(x)}{dx}$$

- Probability mass function (pmf):is a function that gives the probability that a discrete random variable is exactly equal to some value.

$$P_X(x_i) = P(X = x_i)$$

Discrete versus Continuous Random Variables

Discrete Random Variable	Continuous Random Variable
Takes discrete values e.g. {0, 1, 2, 3}	Infinite (continuous) Sample Space e.g. [0,1], [2.1, 5.3]
Probability **Mass** Function (PMF) $p(x_i) = P(X = x_i)$ 1. $p(x_i) \geq 0$, for all i 2. $\sum_{i=1}^{\infty} p(x_i) = 1$	Probability **Density** Function (PDF) $f(x)$ 1. $f(x) \geq 0$, for all x in R_X 2. $\int_{R_X} f(x)dx = 1$ 3. $f(x) = 0$, if x is not in R_X
Cumulative Distribution Function (CDF) $p(X \leq x)$	
$p(X \leq x) = \sum_{x_i \leq x} p(x_i)$	$p(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$ $p(a \leq X \leq b) = \int_a^b f(x)dx$

Statistical Characterizations

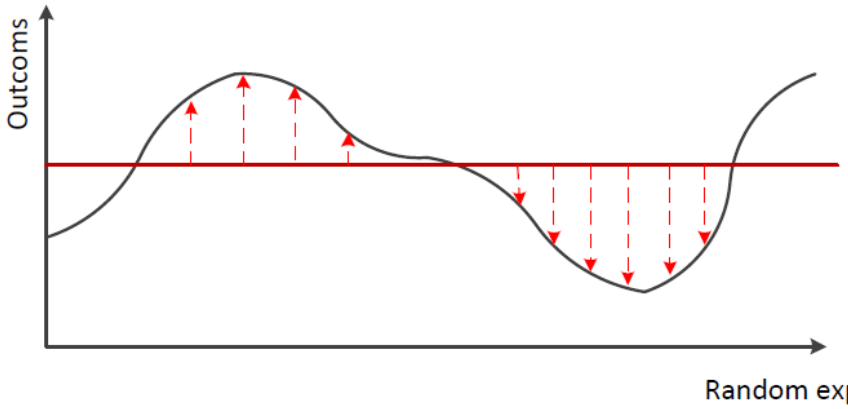
- Expectation (Mean Value) of a random variable

$$\mu_X = E(x) = \int_{-\infty}^{\infty} x f(x)\, dx \quad \text{(for continuouse r.v.)}$$

$$= \sum_i x_i P(X = x_i) \quad \text{(for discrete r.v.)}$$

- Variance of X

$$\sigma_X^2 = E\left[(X - \mu_X)^2\right] = E(X^2) - \mu_X^2$$



Mean characterizes the long tern average amplitude of the r.v.

Variance characterizes how dynamic a random r.v. is

1- Binomial Distribution

Binomial distribution (n, p)

- A fixed number of trials, n, e.g., 15 tosses of a coin;
- A binary outcome, called "success" and "failure", e.g., head or tail in each toss of a coin

  Probability of success is p, probability of failure q = 1 – p
- Constant probability for each observation (independent trials), e.g., probability of getting a tail is the same each time we toss the coin

X is a r. v. that refers to the number of successes in n trials

$$P(X = k) = \binom{n}{k} p^k q^{n-k}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

np = mean value

Example:

- Every packet has *n* bits. There is a probability p_B that a bit gets corrupted. What is the probability that a packet has exactly 1 corrupted bit?

$$P(X=1)=\frac{n!}{1!(n-1)!}p_B^1(1-p_B)^{n-1}=\frac{n!}{(n-1)!}p_B(1-p_B)^{n-1}$$

- What is the probability that the packet is not corrupted?

$$P(X=0)=\frac{n!}{0!(n)!}p_B^0(1-p_B)^n=(1-p_B)^n$$

- What is the probability that the packet is corrupted? At least one bit is corrupted

$$p_E=1-P(X=0)=1-(1-p_B)^n$$

2- Geometric Distribution

- Independent trials - A binary outcome, called "success" and "failure", e.g., head or tail in each toss of a coin

  Probability of success is p, probability of failure q = 1 – p

- Constant probability for each observation (independent trials), e.g., Probability of getting a tail is the same each time we toss the coin
- X is a r. v. that refers to the number of trials needed to obtain the first success

$$P(X = k) = p(1-p)^{k-1}, \quad k = 1, \ldots$$

Example: what is the probability that we need *k* die tossing in order to obtain a head?

Example:

- A wireless network protocol uses a stop-and-wait transmission policy. Each packet has a probability p_E of being corrupted or lost.
- What is the probability that the protocol will need 3 transmissions to send a packet successfully?
  - Solution: 3 transmissions is 2 failures and 1 success, therefore:

$$P(X=3)=(1-p_E)p_E^2$$

- What is the average number of transmissions needed per successful transmission?

$$E[X]=\sum_{i=1}^{\infty} iP(X=i)=\sum_{i=1}^{\infty} i(1-p_E)p_E^{i-1}=$$

$$(1-p_E)\sum_{i=1}^{\infty} ip_E^{i-1}=(1-p_E)\frac{1}{(1-p_E)^2}=\frac{1}{1-p_E}$$

3- Poisson Distribution

1. Poisson distribution defines the probability that a random event occurs k times during a given unit of time interval (e.g., 1 second). λ is a positive real number, equal to the expected number of occurrences that occur during the given interval.
2. A discrete random variable (RV) X follows the Poisson distribution with parameter λ if its probability mass function is:

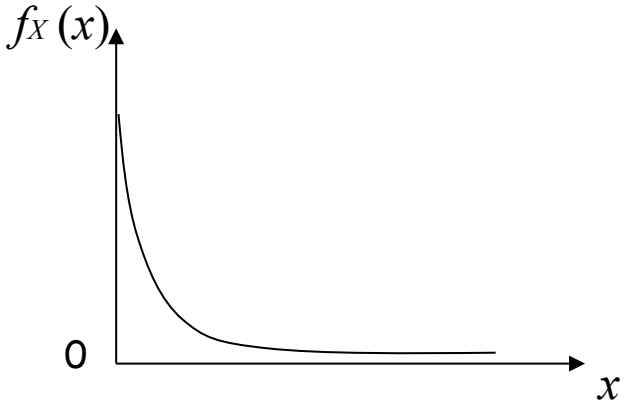
$$P\{X = k\} = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k = 0, 1, 2, 3, \cdots$$

$$E[X] = \lambda, \quad Var(X) = \lambda$$

4- Exponential Distribution

- Exponential distribution : *A continuous RV X follows the exponential distribution with parameter μ,* if its probability density function (PDF) is:

$$f_X(x) = \begin{cases} \mu e^{-\mu x}, & \text{if } x \geq 0; \\ 0, & \text{if } x < 0. \end{cases}$$



- Cumulative distribution function (CDF):

$$F_X(x) = P\{X \leq x\} = \begin{cases} 1 - e^{-\mu x}, & \text{if } x \geq 0; \\ 0, & \text{if } x < 0. \end{cases}$$

- Mean and Variance:

$$E[X] = \frac{1}{\mu}, \quad Var(X) = \frac{1}{\mu^2}$$

4- Exponential Distribution

- Memoryless property: Past history has no influence on the future. The probability of having to wait at least x additional seconds given that one has already waited t seconds.

$$P\{X > x + t | X > t\} = P\{X > x\}$$

Proof: $P\{X > x + t | X > t\} = \dfrac{P\{X > x + t, X > t\}}{P\{X > t\}}$

$$= \frac{P\{X > x + t\}}{P\{X > t\}} = \frac{e^{-\mu(x+t)}}{e^{-\mu t}}$$

$$= e^{-\mu x} = P\{X > x\}$$

- Exponential distribution is the only continuous distribution with the memoryless property

Example

- Let X be the amount of time a customer spends in a bank. X is exponentially distributed with mean 10 minutes.

1) What is the probability that a customer will spend more than 15 minutes in the bank?

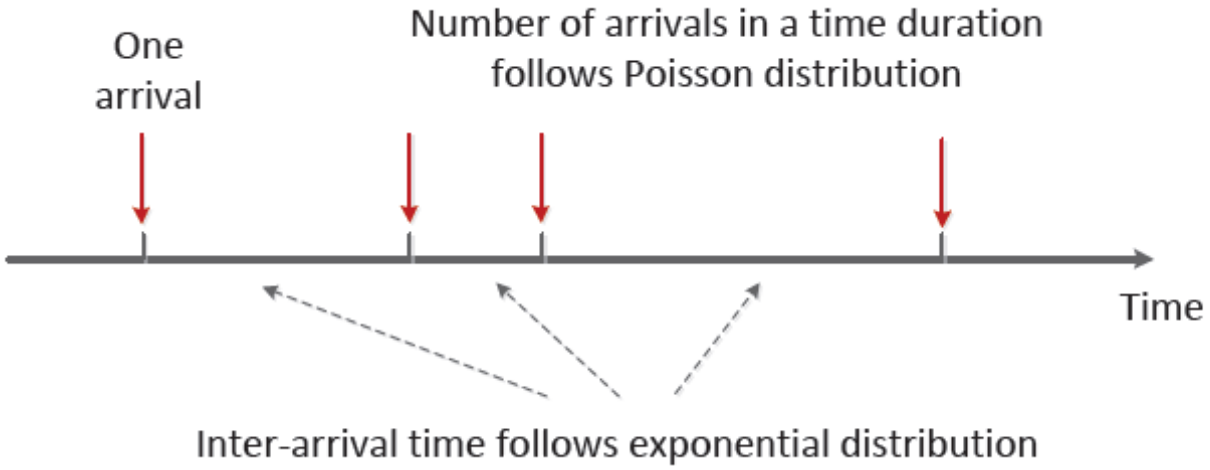
$$\mu = \frac{1}{10} \qquad P(X > 15) = 1 - P(X \leq 15)$$
$$= 1 - (1 - e^{-15\mu}) = e^{-3/2}$$

2) Given that the customer is still in the bank after 10 minutes, what is the probability that he/she will spend more than 15 minutes in the bank?

$$P(X > 5) = e^{-5\mu} = e^{-1/2}$$

5. Poisson Process with rate λ

- Random process : a collection of random variables indexed by a set T ("time").
- Poisson process: A counting process where the number of arrivals in any time interval t follows Poisson distribution with parameter $\lambda * t$, and the **inter-arrival times are independent, identically distributed and follows exponential distribution** with parameter λ



PMF: $$P\{X(t) = k\} = \frac{e^{-\lambda t}(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \ldots$$ *for a time interval of length t*

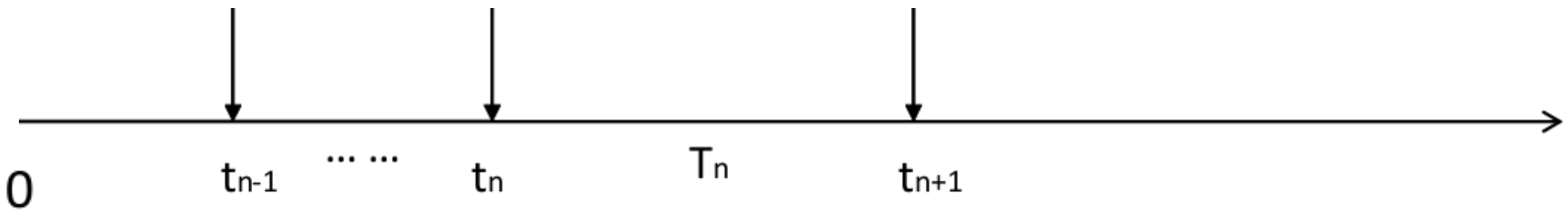
5. Poisson Process with rate λ

- Inter-arrival times for a Poisson process are independent and follow an exponential distribution with parameter λ

Let t_n be the time of n-th arrival;

$\tau_n = t_{n+1} - t_n$ is the n-th inter-arrival time

$$P\{\tau_n \le s\} = 1 - e^{-\lambda s}, \quad s \ge 0$$



$$P\{\tau_n \le s\} = 1 - P\{\tau_n > s\} = 1 - P\{\overbrace{A(t_n + s) - A(t_n) = 0}^{\text{No arrival in the inter-arrival time}}\}$$

$$= 1 - P\{X(s) = 0\} \quad \text{which is the CDF of exponential R.V.}$$

$$= 1 - e^{-\lambda s}\frac{(\lambda s)^0}{0!} = 1 - e^{-\lambda s}$$

Roadmap

- Randomness in communication networks
- A review of probability
- Queuing theory

Queuing theory

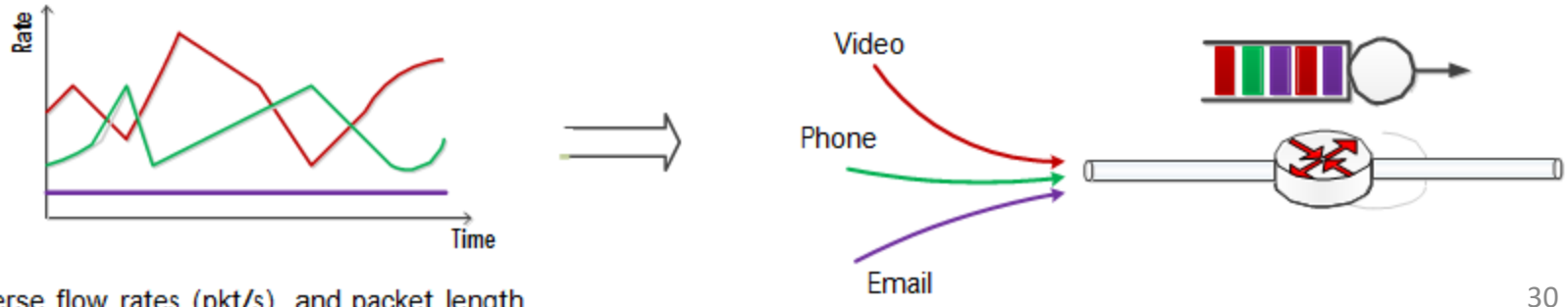
- Introduction to queuing theory
  - What is queuing theory?
  - Why study queues?
  - Notations of queues
- Queuing Models
  - M/M/1 Queue
  - M/M/1/N Queue
  - Network of queues

What is queuing theory used for?

- A random variable describes the behavior of one random event

  e.g., User arrivals in unit time

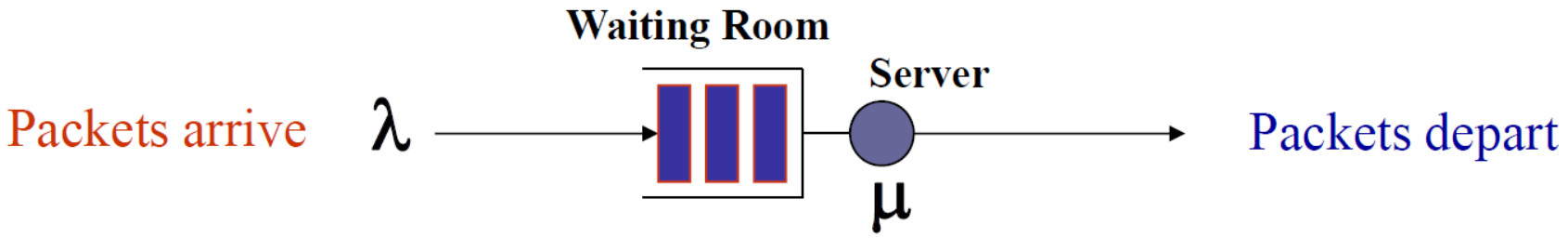
- How to model and analyze **a system** where multiple random events co-exist?
  - e.g., at a router, multiple flows with different rates are injected
- Queuing theory: a collection of mathematical models of various queuing systems



Diverse flow rates (pkt/s), and packet length

Queuing Representation

1- Single Server Queuing Model

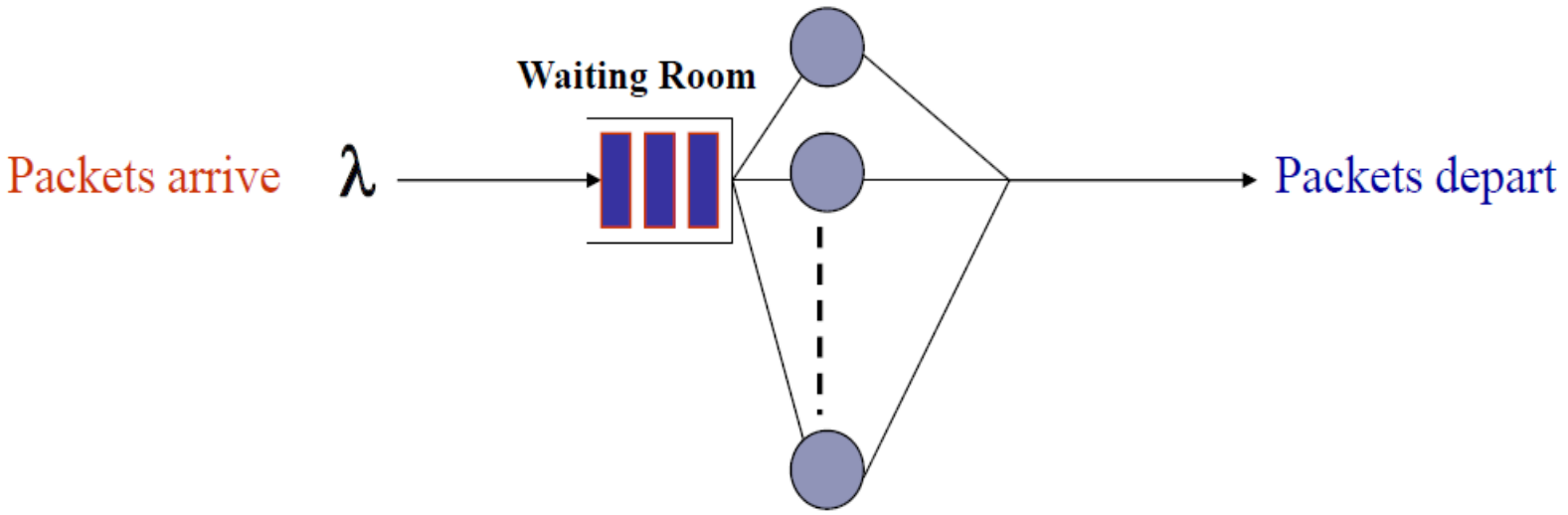


- The server is usually the transmission facility
- The arrivals and the service time are random.
- The service time: how long a packet will remain in service, directly related to the length of a packet
- The buffer is the available space in the system.

Queuing Representation

2- Multi-Server Queue

Arriving packets first go to empty servers. If more than one server is empty, any server is chosen randomly.

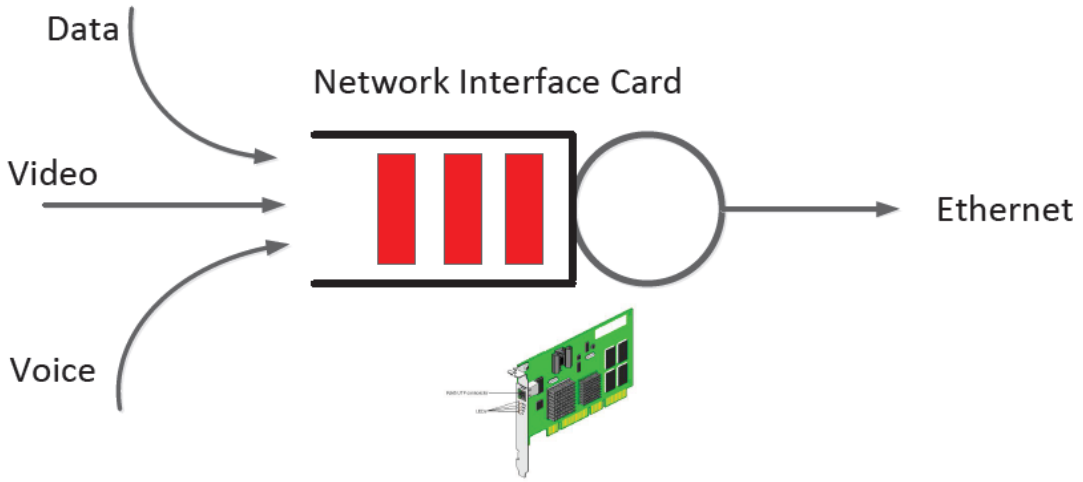


Queue is everywhere

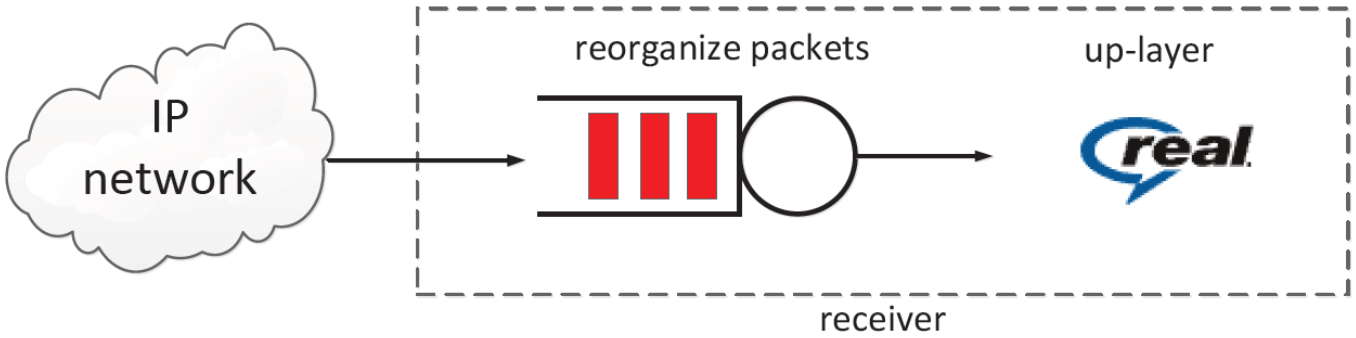
Packet switching relies on queues

- Queues are everywhere

At the sender

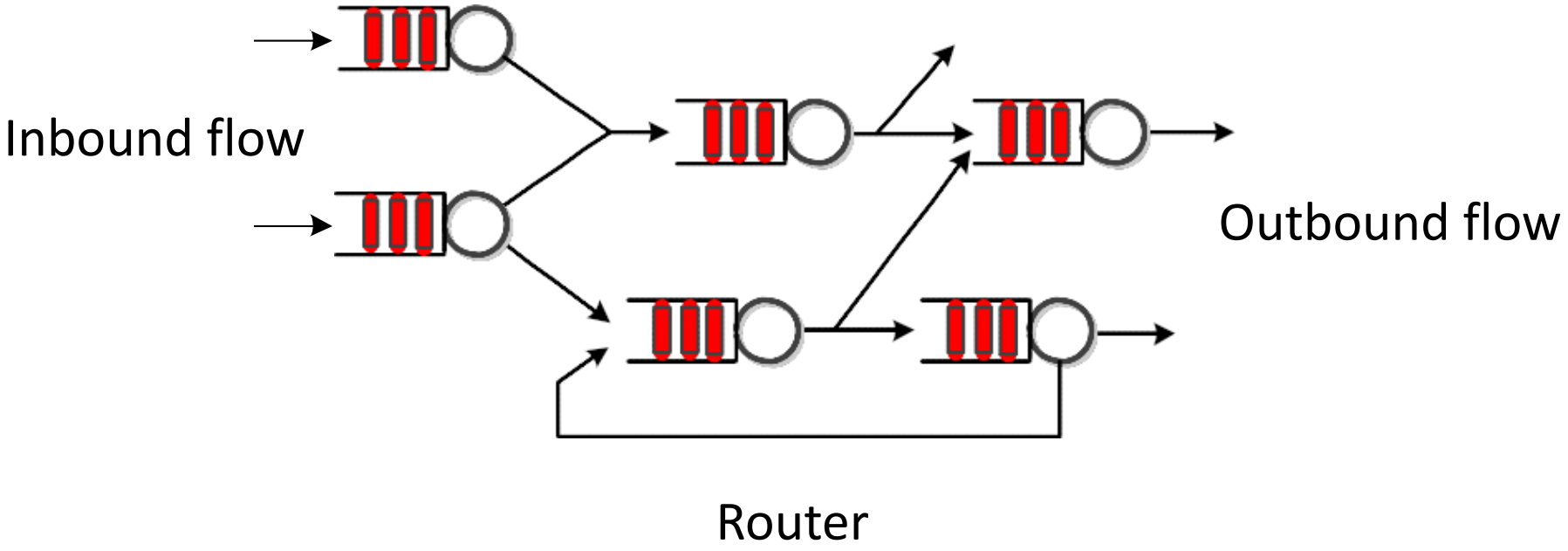


At the receiver



Queue is everywhere

- A computer network is a network of queues!



Goals

- Evaluate the system performance
  - Average queue length
  - Average waiting time
  - Loss probability due to buffer overflow
- Improve the system performance
  - Provide guaranteed packet loss probability with large enough buffer (How large is enough?)
  - Control the queue to offer differentiated service to users

Queues are represented via the Kendal notation: A/S/C/K

- A: The arrival process of packets. M stands for Markovian (Poisson) process; λ is the arrival rate. The interarrival time is exponentially distributed with mean 1/λ.

- S: The packet departure process. In case of M, the interdeparture time (the service time) is exponentially distributed with average service time 1/μ, where μ is the service rate.

- C: the number of parallel servers in the system.

- K: max. number of packets in the system (the max. number of packets that can be accommodated in the buffer plus the number of servers). If K is missing, K = infinity.

Examples

- M/M/1:
  - Poisson arrivals (Markovian— a process with no memory), Poisson departure process, 1 server, infinite capacity and population
  - The simplistic ”realistic” queue
- M/M/1/N
  - Same as above, but the capacity is N
- M/M/m/m
  - *m* parallel servers, system capacity is m
- M/M/m
  - *m* parallel servers, Infinite capacity
- G/G/1
  - General arrival and departure (service) distributions, 1 server

Queuing theory

- Introduction to queuing theory
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A simple queueing model (M/M/1 queue)



1 server, infinite buffer size

Assumptions:

1) Customers arrive according to a Poisson process with parameter λt , where λ is the average number of arrivals per unit time (arrival rate).

Let *X(t)* be the number of arrival in [0 , t] ,

$$P(X(t)= k) = (\lambda t)^k \cdot e^{-\lambda t}/k!, \quad k = 0, 1, 2, \cdots$$

A simple queueing model (M/M/1 queue)

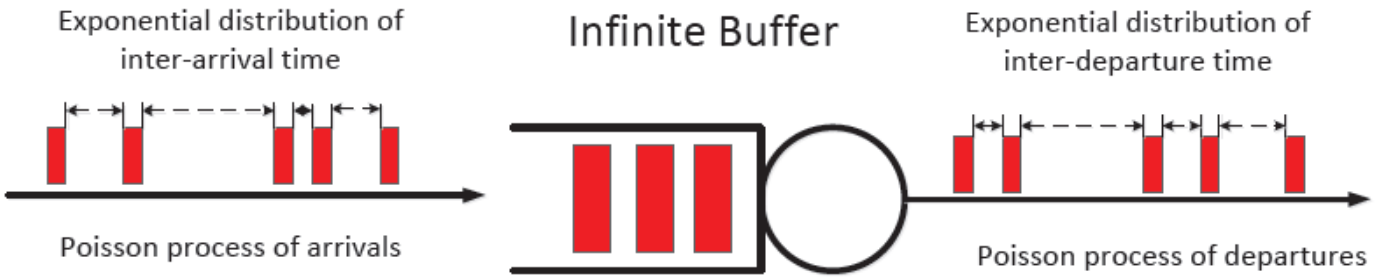
(2) The single server takes a random length of time, T_s, to serve each customer and these times are independent random variables for different customers.

T_s has an exponential distribution with parameter μ :

$$f_{T_s}(t) = \mu e^{-\mu t}, \quad t \geq 0$$

$$F_{T_s}(t) = 1 - e^{-\mu t}, \quad t \geq 0$$

where μ is the average number of customers being served per unit time (departure rate).



A simple queueing model (M/M/1 queue)

Suppose we observe an arrival at t and the next arrival at $t+\tau$, where τ is a random variable, then

$$\begin{aligned} P(\tau > x) &= P(\text{no arrival occurs in } (t, t+x)) \\ &= P(X = 0 \text{ in } (t, t+x)) \\ &= e^{-\lambda x}, \quad \tau > 0, x \geq 0 \end{aligned}$$

$$\Rightarrow F_\tau(x) = P(\tau \leq x) = 1 - P(\tau > x) = 1 - e^{-\lambda x}, x \geq 0$$

$$\Rightarrow f_\tau(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

i.e. τ has an exponential distribution with parameter λ.

A simple queueing model (M/M/1 queue)

Property of the exponential distribution

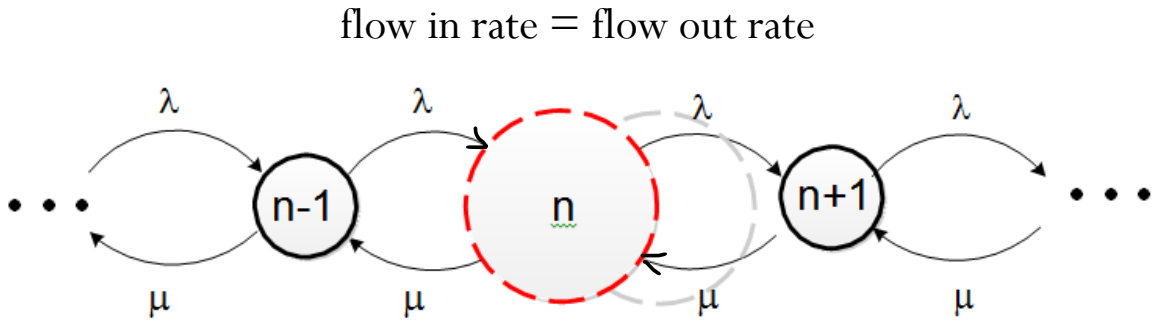
$$
\begin{aligned}
P(t \le T_s \le t + \Delta t | T_s \ge t) &= \frac{P(t \le T_s \le t + \Delta t)}{P(T_s \ge t)} \\
&= \frac{F_{T_s}(t + \Delta t) - F_{T_s}(t)}{1 - F_{T_s}(t)} \\
&= \frac{e^{-\mu t}(1 - e^{-\mu \Delta t})}{e^{-\mu t}} \\
(\text{ as } \Delta t \to 0) \quad &= 1 - e^{-\mu \Delta t} \approx \mu \Delta t
\end{aligned}
$$

i.e., the probability of the completion of a service in the next Δt seconds is a constant, independent of how long the service has been going on! The service time has no memory.

Both random variables τ and T_s have no memory!

Summary: A single server queue with Poisson arrival and exponential service times is denoted as M/M/1 queue (where M stands for Markovian— a process with no memory).

M/M/1: State Balance Equation



- P_n is the steady state probability at state n
  - Draw the transition diagram
  - Write balance equation:

$$\begin{cases} (\lambda+\mu)P_n = \lambda P_{n-1} + \mu P_{n+1}, n \geq 1 \\ \lambda P_0 = \mu P_1 \qquad \text{(balance condition)} \end{cases}$$

  - At any state: rate at which the process leaves = rate at which it enters
  - Calculate P_n

M/M/1 queue

- Find P_0:

  Since $\sum_{i=0}^{\infty} P_i = 1$ we have

  $$P_0(1 + \rho + \rho^2 + \cdots) = P_0 \cdot \frac{1}{1-\rho} = 1, \text{ if } 0 < \rho < 1$$

  Then, $P_0 = 1 - \rho$,

  $$P_n = P_0 \cdot \rho^n = (1 - \rho) \cdot \rho^n, \quad n = 0, 1, 2, \cdots,$$

- The **average number of customers** in the system is:

$$L = E(N) = \sum_{n=0}^{\infty} nP_n = \sum_{n=0}^{\infty} n(1-\rho)\rho^n = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1-\rho}$$

- The **average number of customers in the queue** is:

$$E(N_q) = \sum_{n=1}^{\infty} (n-1)P_n = \sum_{n=1}^{\infty} (n-1)(1-\rho)\rho^n = \frac{\rho^2}{1-\rho} = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

Little's Law

- The long-term average number of customers in the system is

$$E(N) = \lambda W$$

where $E(N)$ is the average number of customers in the system
λ is the average arrival rate of customers
W is the average time a customer spends in the system.
(W= time wait in the buffer + time receiving service)

- The long-term average number of customers in the buffer/ queue is

$$E(N_q) = \lambda T_q$$

where $E(N_q)$ is the average number of customers in the buffer/ queue
T_q is the average time a customer spends in the buffer/ queue

M/M/1 queue

$$E(N_q) = \frac{\rho^2}{1-\rho} = \frac{\lambda^2}{\mu(\mu-\lambda)} \qquad L = E(N) = \frac{\lambda}{\mu-\lambda} = \frac{\rho}{1-\rho}$$

- Let T_q denote the mean time that a customer has to wait in the queue before getting service

$$T_q = \frac{\rho}{\mu(1-\rho)} = \frac{\lambda}{\mu(\mu-\lambda)}, \quad \mu > \lambda$$ (By little's law)

- Let W denote the mean time that a customer has to wait from the moment he arrives until he departures then

$$W = \frac{\lambda}{\mu-\lambda}/\lambda = \frac{1}{\mu-\lambda}$$ (By little's law)

$$W = T_q + T_s = T_q + \frac{1}{\mu} = \frac{\lambda}{\mu(\mu-\lambda)} + \frac{1}{\mu} = \frac{1}{\mu-\lambda}$$

M/M/1 queue

- Example: In an M/M/1 queue, customers arrive at the rate of λ =15 per hour. What is the minimum server rate to ensure that

1. the sever is idle at least 10% of the time

$$\text{For } P_0 \geq 0.10 \Rightarrow 1-\rho \geq 0.10 \Rightarrow \frac{\lambda}{\mu} \leq 0.9 \Rightarrow \mu \geq 16.67$$

2. the expected value of the queue length is not to exceed 10?

$$E(N_q) = \frac{\rho^2}{1-\rho} = \frac{\lambda^2}{\mu(\mu-\lambda)} \leq 10 \Rightarrow \mu = 16.375$$

3. the probability that at least 20 people in the queue is at most 50% .

$$P_n = (1-\rho)\rho^n,\ n = 0, 1, 2, \cdots,$$
$$P(n \geq 21) = \sum_{n=21}^{\infty}(1-\rho)\rho^n \leq 0.5 \Rightarrow \mu \geq 15/e^{ln(0.5)/21}.$$

Note: for that at least 20 people in the queue, there are at least 21 people in the system.

M/M/1 queue

- Example 2.2: Consider a packet transmission system whose arrival rate (in packet/sec) is kλ, (k> 1), and departure rate is kμ (service time is 1/(kμ)). What is the average number of packets in the system? what is the average total delay per packet?

- Solution: i)

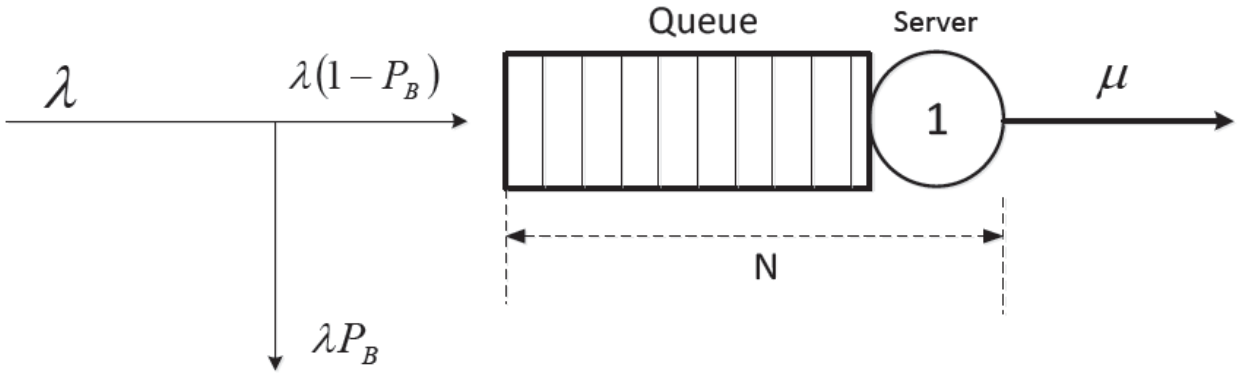
$$L = \frac{k\lambda}{k\mu - k\lambda} = \frac{\lambda}{\mu - \lambda}$$

- ii) according to the Little's Law, the average total delay per packet is

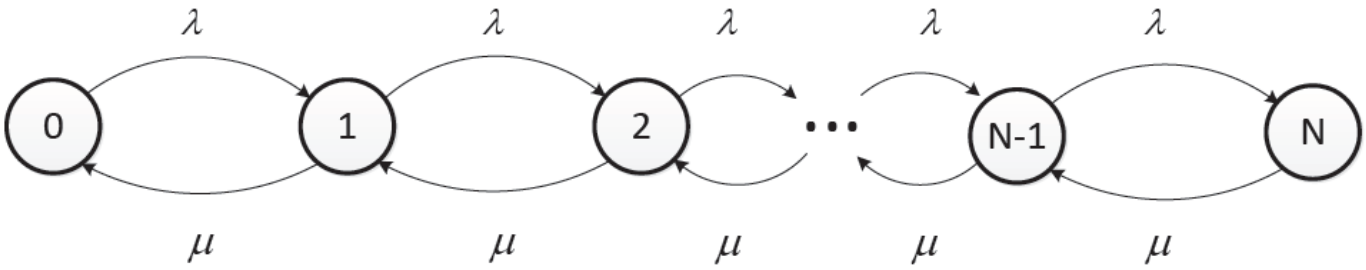
$$W = \frac{L}{k\lambda} = \frac{1}{k(\mu - \lambda)}$$

(Increasing the arrival and transmission rates by the same factor, the average total delay is reduced by the factor)

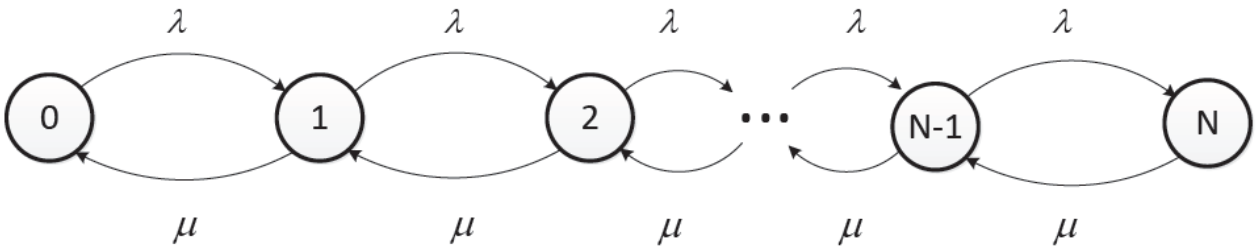
M/M/1/N: Finite Buffer Queue



  - Arrival packets are dropped when the buffer is full
  - P_B: blocking probability (prob. that buffer is full)
- State transition diagram



M/M/1/N: Blocking Probability P_B



- According to the state balance equation, we have

$$P_n = \rho^n P_0 \quad 0 \le n \le N$$

- As the sum of probability is one (axiom of probability),

$$\sum_0^N P_n = 1 \Rightarrow P_0 \sum_{n=0}^{\infty} \rho^n = P_0 \frac{1-\rho^{N+1}}{1-\rho} = 1$$

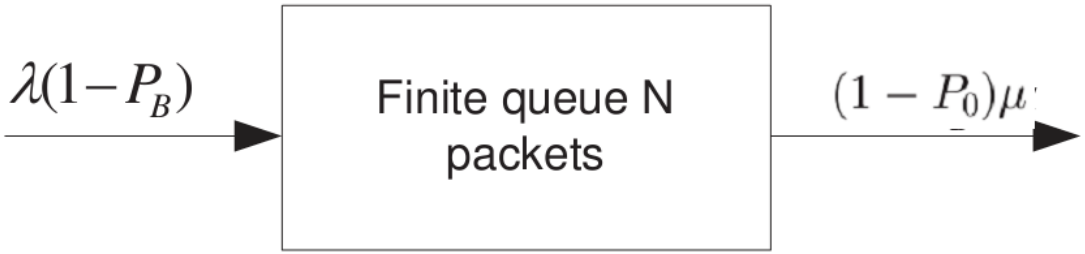
- i.e.,

$$P_0 = \frac{1-\rho}{1-\rho^{N+1}} \; and \; P_n = P_0\rho^n = \frac{1-\rho}{1-\rho^{N+1}}\rho^n$$

- Therefore, the blocking probability is

$$P_B = P_N = P_0\rho^N = \frac{1-\rho}{1-\rho^{N+1}}\rho^N$$

M/M/1/N queue



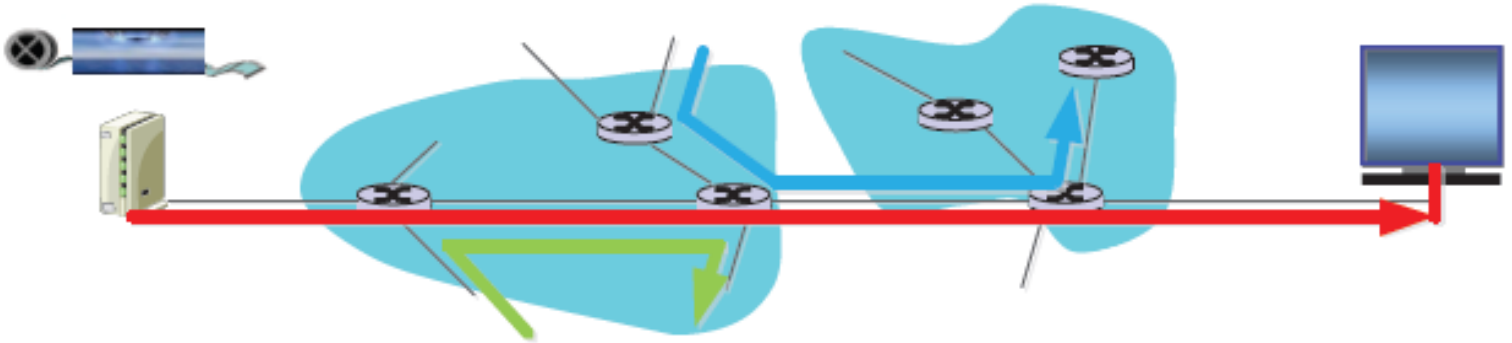
Arrival rate: λ packet/sec;

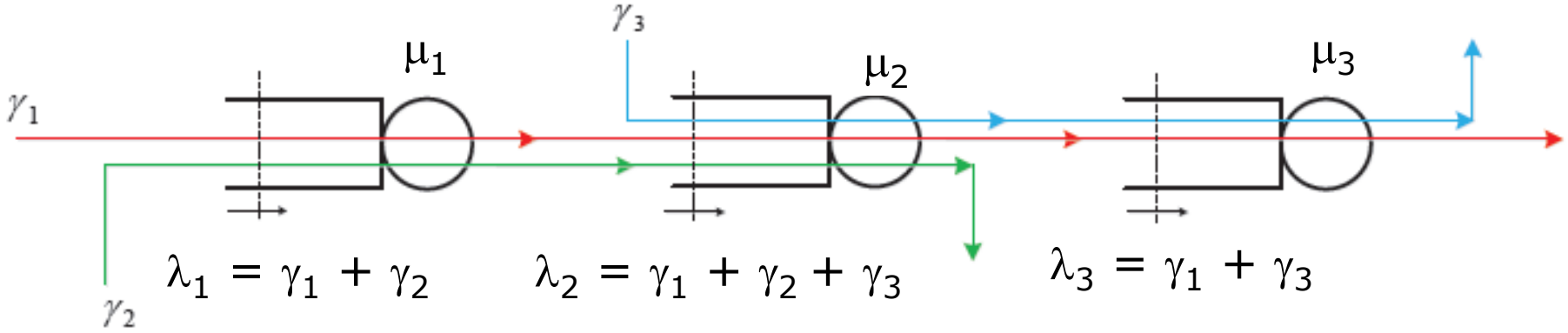
Blocking probability: P_B ;

Departure rate: μ packet/sec;

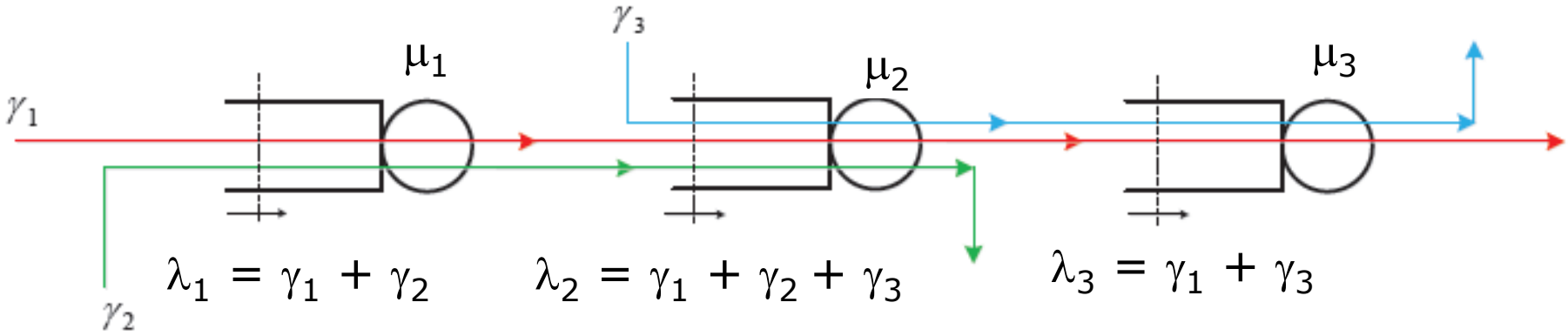
Throughput: $\lambda(1-P_B) = (1-P_0)\mu$

Network of M/M/1 Queues





- Each router i is modeled as a M/M/1 queue with packet arrival rate λ_i and departure rate μ_i
- Assuming the packet arrivals and departures at routers are independent
- Assuming L flows γ_1, γ_2 and γ_3 are the packet rates of traffic flows

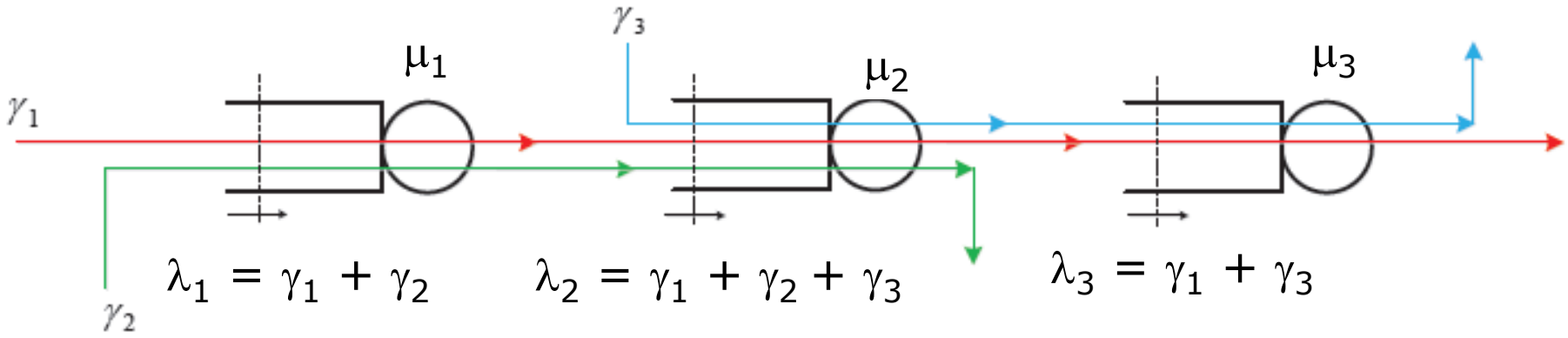


- L tandem M/M/1 queues (routers)
- The average number of customers in ith M/M/1 queue

$$E(N_i) = \frac{\rho_i}{1-\rho_i} = \frac{\lambda_i}{\mu_i - \lambda_i}$$

- The total number of customers in the system

$$E(N) = \sum_{i=1}^{L} E(N_i) = \sum_{i=1}^{L} \frac{\lambda_i}{\mu_i - \lambda_i}$$



- The average sojourn time of packet in the system (using Little's Law)

$$E(T) = \frac{E(N)}{\lambda} = \frac{1}{\lambda} \sum_{i=1}^{L} \frac{\lambda_i}{\mu_i - \lambda_i}$$

where λ is the average arrival rate of the customers entering the system

The average service time through the system is a weighted sum of the average service time through each queuing component.